ISIC 2024 Skin Cancer Detection: Multi-Modal Classification with GBDT Ensemble, Deep Learning, and Stacking Meta-Learner

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***Abstract*—We present a three-phase machine learning pipeline for binary classification of malignant versus benign skin lesions using the ISIC 2024 SLICE-3D dataset [1]. The dataset contains 401,059 lesion records from 1,042 patients with 393 malignant cases (0.098%), posing extreme class imbalance and patient-group leakage challenges. Phase 1 (tabular) combines clinically motivated ABCDE feature engineering, patient-wise ugly duckling z-scores, and a GBDT ensemble (LightGBM, XGBoost, CatBoost) with isotonic calibration, achieving pAUC = 0.1672. Phase 2 (image) fine-tunes four pretrained vision backbones (EfficientNetV2-S, EVA02-Small, ConvNeXtV2-B, SwinV2-B) on 401K lesion images using focal loss, weighted sampling, and test-time augmentation (TTA) with D4 symmetry transforms; the best individual image model (SwinV2-B) achieves pAUC = 0.1549. Phase 3 (stacking) combines all five out-of-fold (OOF) prediction sources through rank transformation and a logistic regression meta-learner, achieving a final pAUC = 0.1745 — a 5.6% improvement over the best individual model. All phases share identical cross-validation splits (`StratifiedGroupKFold`, k=5, grouped by patient) to ensure leak-free OOF alignment for stacking.**

I. Introduction

Melanoma is responsible for the vast majority of skin cancer deaths despite comprising only ∼1% of diagnoses [11]. Five-year survival drops from 98% (localised) to 32% (stage IV), making early detection critical. The ISIC 2024 SLICE-3D challenge [1] provides both large-scale structured lesion measurements extracted from Canfield's Vectra Total Body Photography (TBP) system and dermoscopic images, enabling multi-modal approaches.

This paper describes a three-phase pipeline: (i) tabular GBDT ensemble on structured metadata, (ii) fine-tuned pretrained vision backbones on lesion images, and (iii) a stacking meta-learner that fuses predictions from both modalities. The primary metric is partial AUC above 80% TPR (pAUC, range $[0, 0.2]$).

The `tbp_lv_*` column family is not arbitrary metadata: it numerically encodes the canonical ABCDE clinical criteria [12] as machine-extracted measurements (Table I), making the metadata directly actionable for melanoma screening.

Contributions:

1) A rigorous leakage audit removing columns that inflate training pAUC to 0.197 yet fail at inference.
2) A staged feature pipeline (55 → 45 → 58 → 271 → 88 columns) with ABCDE hand-crafted features and ugly duckling patient-relative z-scores.

TABLE I: ABCDE Criteria Mapping to Dataset Columns.

Criterion	Clinical Meaning	Key Columns
Asymmetry	Irregular shape	`symm_2axis`, `eccentricity`
Border	Ragged edge	`norm_border`, `perimeterMM`
Color	Colour contrast	`deltaLBnorm`, `color_std_mean`
Diameter	Large size	`long_diam_mm`, `areaMM2`
Evolution	Change / AI score	`nevi_confidence`, `dnn_conf`

3) Diagnosis and fix of LightGBM's count-based `min_child_samples` under-performing with `scale_pos_weight`, raising pAUC from 0.1217 to 0.1400.
4) A multi-architecture image model sweep across four backbone families (CNN, ViT, ConvNet, hybrid) with focal loss and D4 TTA.
5) A stacking meta-learner achieving pAUC = 0.1745 by fusing tabular and image signals through rank-transformed features.

II. Related Work

a) ISIC Challenges: The ISIC has hosted annual dermoscopy classification challenges since 2016 [14]. While earlier editions focused on CNNs for image classification, the 2024 edition introduced TBP metadata, enabling tabular ML approaches. Top solutions combined image and tabular features through stacking or late fusion.

b) GBDTs for Tabular Data: GBDTs remain the dominant paradigm for structured data [13], consistently outperforming deep learning on medium-sized tabular benchmarks. LightGBM [2], XGBoost [3], and CatBoost [4] offer complementary inductive biases (leaf-wise vs. level-wise vs. ordered boosting).

c) Transfer Learning for Medical Imaging: Fine-tuning ImageNet-pretrained backbones is the dominant paradigm for medical image classification [26]. EfficientNetV2 [17] introduced progressive training and fused-MBConv blocks for efficiency. Vision Transformers (ViT) [18] and their variants — EVA02 [19] with masked image modelling pre-training, SwinV2 [20] with shifted window attention, and ConvNeXtV2 [21] modernising pure convolutions — have achieved state-of-the-art results across vision benchmarks. For skin lesion classification specifically, ensemble diversity across architecture families (CNN, ViT, hybrid) is more valuable than single-model accuracy [27].

TABLE II: ISIC 2024 SLICE-3D Training Set.

Property	Value
Total lesion records	401,059
Unique patients	1,042
Avg. lesions / patient	≈ 385
Raw metadata columns	55
Image format	JPEG in HDF5
Malignant (target = 1)	393 (0.098%)
Benign (target = 0)	400,666 (99.9%)
Class ratio (neg:pos)	$\approx 1020 : 1$

TABLE III: Leakage Audit: Columns Excluded from Features.

Column(s)	Reason
`mel_thick_mm`, `mel_mitotic_index`	Only populated for confirmed malignant cases.
`iddx_full`, `iddx_1–iddx_5`	Diagnostic label strings; direct target leakage.
`lesion_id` → `has_lesion_id`	All 393 malignant rows have `lesion_id`; test set lacks the column entirely.
`isic_id`, `image_type`, `copyright_license`	Zero variance / administrative.

d) Focal Loss: Lin et al. [22] proposed focal loss to down-weight easy negatives in object detection. For binary classification with extreme imbalance, focal loss modulates the standard cross-entropy:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

where γ controls focus on hard examples and α_t balances classes. We use γ=2, α=0.25.

e) Probability Calibration: Modern tree ensembles often produce poorly calibrated probabilities [7]. Isotonic regression [15] is non-parametric and monotone; Platt scaling [6] applies a sigmoid fit; temperature scaling [16] divides logits by a learned scalar T.

f) Stacking and Model Fusion: Stacking [28] trains a meta-learner on out-of-fold predictions from base models. Rank-averaging converts heterogeneous probability scales to a common percentile space, preserving rank-based metrics like AUC [29]. For extremely low positive counts, simple linear meta-learners outperform tree-based stackers due to lower overfitting risk.

III. Dataset and Problem Formulation

A. Dataset Statistics

Table II summarises the training set. The 1:1020 class ratio is the defining challenge: standard accuracy and full-curve AUC both reward predicting benign for every sample. Images are stored as JPEG bytes in HDF5, decoded on-the-fly during training.

B. Competition Metric: Partial AUC

The official scoring metric is the partial area under the ROC curve above a minimum TPR threshold $\tau = 0.80$:

$$\text{pAUC} = \int_{\tau}^{1} \frac{d\,\text{TPR}}{d\,\text{FPR}}\, d\,\text{TPR}, \qquad \text{pAUC} \in [0,\, 1-\tau] \tag{2}$$

The Kaggle implementation uses label-flipping and prediction negation:

$$\text{pAUC}(y, \hat{p}) = \text{AUC}(1-y,\ -\hat{p},\ \text{max_fpr} = 1-\tau) \tag{3}$$

This differs from `sklearn`'s `roc_auc_score(max_fpr=0.2)`, which applies the McClish normalisation [10] and yields systematically different values.

C. Cross-Validation Strategy

All three phases use identical `StratifiedGroupKFold` (k=5, `seed=42`) [9] with groups = `patient_id`. No patient appears in both training and validation within a fold, preventing group leakage from the ≈385 lesions per patient. This shared CV strategy ensures that OOF predictions from all phases are aligned by `isic_id` for leak-free stacking in Phase 3.

D. Leakage Audit

Table III lists columns removed before modelling. The subtlest case is `has_lesion_id`: all 393 malignant training rows have a non-null `lesion_id` (biopsied lesions), giving pAUC = 0.197 in training. The test set contains no `lesion_id` column, however, so the model would output near-zero probability for every test row — a classic leakage trap.

IV. Phase 1: Tabular Feature Engineering

Fig. 1 illustrates the four-stage pipeline. Each stage is fitted independently per CV fold; no test-set statistics influence training.

A. Preprocessing (55 → 45 Columns)

The preprocessor: (i) removes 12 columns (Table III), (ii) label-encodes five categoricals (`sex`, `anatom_site_general`, `attribution`, `tbp_tile_type`, `tbp_lv_location_simple`), (iii) median-imputes `age_approx` (1.7% missing) and adds an `age_approx_missing` indicator, and (iv) fills remaining NaN with 0.

B. ABCDE Feature Engineering (45 → 58 Columns)

Thirteen domain-motivated features are added. Selected equations:

Colour (C) — CIELAB distance and chroma:

$$\Delta E = \sqrt{(\Delta A)^2 + (\Delta B)^2 + (\Delta L)^2} \tag{4}$$

$$\text{chroma} = \sqrt{A_{\text{lesion}}^2 + B_{\text{lesion}}^2} \tag{5}$$

Asymmetry / Border (A, B) — shape descriptors:

$$\text{circularity} = \frac{4\pi \cdot \text{area}}{\text{perimeter}^2 + \varepsilon} \tag{6}$$

$$\text{border_complexity} = \frac{\text{perimeter}}{\sqrt{\text{area}} + \varepsilon} \tag{7}$$

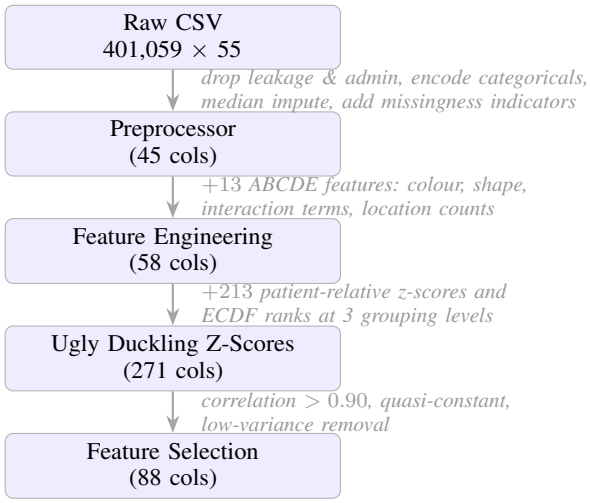

Fig. 1: Four-stage feature pipeline. Each stage is refitted per CV fold so no validation statistics leak into features.

where $\varepsilon = 10^{-8}$ guards against zero denominators.

Interaction features:

$$\text{age_x_area} = \text{age} \times \text{areaMM}^2, \quad \text{ecc} \times \Delta E$$

C. Ugly Duckling Features (58 → 271 Columns)

The *ugly duckling sign* [8] states that a mole markedly different from a patient's other lesions warrants biopsy. For each feature f and grouping G:

$$z_G(f) = \frac{f - \mu_G(f)}{\sigma_G(f) + \varepsilon}, \qquad r_G(f) = \text{ECDF}_G(f) \tag{8}$$

Singleton groups set $z_G = 0$ and $r_G = 0.5$ (centred, neutral). Three grouping levels from the 2nd-place competition solution:

Group	Clinical question
`[patient_id]`	Weird vs. all patient moles?
`[patient_id, site]`	Weird for this body region?
`[patient_id, location]`	Weird for this TBP location?

Three aggregates are also computed: ud_outlier_count $= \sum_f \mathbf{1}[|z_{\text{pat}}(f)| > 2]$, mean absolute z-score, and a single-lesion-patient flag.

D. Feature Selection (271 → 88 Columns)

Three sequential filters: (1) exclusion list (`patient_id`, `target`, `has_lesion_id`), (2) quasi-constant filter (single value $> 99.5\%$ of rows), and (3) correlation filter ($|\rho| > 0.90$; threshold validated at 0.85/0.90/0.95 in EDA notebook, with 0.90 selected).

V. Phase 1: Tabular Models

A. Seed Averaging

Each GBDT is trained with seeds $\{42, 123, 456, 789, 2024\}$ and predictions are averaged per fold, reducing variance from stochastic gradient descent paths.

B. LightGBM [2]

Key hyperparameters: `num_leaves=63`, `max_depth=6`, `learning_rate=0.05`, up to 2000 trees with 100-round early stopping, $L_1 = L_2 = 0.1$, `scale_pos_weight=100`.

a) Critical imbalance fix: `min_child_samples` is a *count-based* constraint that ignores `scale_pos_weight`: with ≈314 positives per fold and `min_child_samples=80`, LightGBM can form at most $\lfloor 314/80 \rfloor = 3$ pure positive leaves. XGBoost's `min_child_weight` is *hessian-sum based*: under logistic loss with `scale_pos_weight=100`, a single positive sample contributes ≈ 25 to the hessian sum, easily satisfying `min_child_weight=10`.

The fix sets:

$$\texttt{min_sum_hessian_in_leaf} = 10.0 \tag{9}$$

This raised pAUC from 0.1217 to **0.1400** and halved the standard deviation (0.0132 → 0.0052).

C. XGBoost [3]

`max_depth=6`, `min_child_weight=10`, `subsample=colsample_bytree=0.8`, $\alpha = \lambda = 0.1$, `scale_pos_weight=100`. The `max_delta_step=1` parameter clips leaf weight updates to $[-1, +1]$, preventing divergence under extreme imbalance.

D. CatBoost [4]

`depth=6`, `learning_rate=0.05`, `l2_leaf_reg=3.0`, up to 2000 iterations with early stopping. Class imbalance is handled via `auto_class_weights=SqrtBalanced` ($\sqrt{N_{\text{neg}}/N_{\text{pos}}}$ weighting).

E. SVM Baseline (Course Requirement) [5]

An SVM with RBF kernel satisfies the course requirement for a classical probabilistic model. Pipeline: `StandardScaler` → `SVC(probability=True, class_weight=balanced)`, where

TABLE IV: Phase 2 Backbone Architectures.

Backbone	Family	Params	Input
EfficientNetV2-S	CNN (MBConv)	20M	224^2
EVA02-Small	ViT (MIM pretrain)	22M	336^2
ConvNeXtV2-B	Pure ConvNet	88M	224^2
SwinV2-B	Shifted-window ViT	87M	256^2

`probability=True` enables Platt scaling [6]. Training uses a 20K stratified subsample per fold ($\mathcal{O}(n^3)$ prohibitive at full scale). The SVM is excluded from the final ensemble due to high variance (pAUC std $= 0.0245$).

F. Rank-Averaging Ensemble

Rank-averaging is robust to probability scale differences across models:

$$\hat{r}_m = \frac{\text{rank}(\hat{p}_m)}{N+1}, \qquad \hat{p}_{\text{ens}} = \frac{1}{3}\sum_{m=1}^{3} \hat{r}_m \tag{10}$$

where the sum is over LightGBM, XGBoost, and CatBoost (SVM excluded).

G. Isotonic Calibration [7]

Isotonic regression is fitted on the full set of OOF ensemble predictions. As a monotone transformation, calibration preserves score rankings and therefore cannot decrease pAUC.

VI. Phase 2: Image Deep Learning

Phase 2 fine-tunes pretrained vision backbones on 401K lesion images to capture visual patterns (texture, border irregularity, colour heterogeneity) not represented in the tabular metadata.

A. Architecture

We use the `timm` library [23] to load ImageNet-pretrained backbones. The classification head replaces the original with: `Dropout`(0.3) $\rightarrow$ `Linear`(d, 1), where d is the backbone's feature dimension. The sigmoid output produces a malignancy probability.

Table IV lists the four backbones selected to maximise architectural diversity:

a) Architecture selection rationale: The four backbones span three distinct inductive bias families: (1) **EfficientNetV2-S** [17] uses fused-MBConv blocks with strong local spatial priors; (2) **EVA02-Small** [19] is a plain ViT pretrained with masked image modelling (MIM), providing global self-attention from the first layer; (3) **ConvNeXtV2-B** [21] modernises pure convolutions with FCMAE pretraining and Global Response Normalisation; (4) **SwinV2-B** [20] uses shifted-window attention with log-spaced continuous position bias, combining local and global reasoning hierarchically. This diversity ensures the image models make *different errors*, which is critical for stacking to improve over any individual model [29].

B. Training Configuration

a) Loss function: Focal loss (Eq. 1) with γ=2, α=0.25 down-weights the overwhelming easy-negative majority, focusing gradient updates on hard boundary cases and rare positives.

b) Sampling: `WeightedRandomSampler` with neg:pos ratio of 50:1 ensures each mini-batch contains approximately 2% positive samples, compared to the natural 0.098%. This is less aggressive than full balancing (which would oversample positives $\sim$1000$\times$) and reduces overfitting to the minority class.

c) Optimiser: AdamW [24] with differential learning rates: the pretrained backbone uses $\text{lr}_{\text{backbone}} = 5 \times 10^{-5}$ (slow fine-tuning to preserve learned representations), while the randomly initialised classification head uses $\text{lr}_{\text{head}} = 5 \times 10^{-4}$ (10$\times$ faster to converge quickly). Weight decay is 10^{-4}.

d) Learning rate schedule: OneCycleLR [25] with cosine annealing ramps the learning rate from $\text{lr}/25$ to the peak, then decays to $\text{lr}/10^4$. This warm-up phase is critical for large pretrained models to avoid destructive early gradient steps.

e) Data augmentation: Training augmentations: horizontal flip, vertical flip, 90° rotation, transpose, colour jitter (brightness/contrast $\pm 20\%$, saturation/hue $\pm 10\%$), Gaussian blur ($\sigma \in [0.1, 2.0]$), and coarse dropout (max 8 holes, 8×8 pixels). All augmentations respect the D4 dihedral symmetry group of lesion images (no preferred orientation).

f) GPU strategy: Smaller models (EfficientNetV2-S, EVA02-Small, 20–22M params) train on NVIDIA T4 (16 GB), while larger models (ConvNeXtV2-B, SwinV2-B, 87–88M params) require L40S (48 GB). Gradient accumulation (4$\times$) is used to simulate larger effective batch sizes when GPU memory is constrained.

C. Test-Time Augmentation (TTA)

At inference, each image is transformed by all 8 elements of the D4 dihedral group (4 rotations $\times$ 2 reflections). Predictions are averaged in sigmoid (probability) space:

$$\hat{p}_{\text{TTA}} = \frac{1}{8}\sum_{g \in D_4} \sigma(f_\theta(g \cdot x)) \tag{11}$$

This exploits the fact that lesion orientation is arbitrary in dermoscopy, and averaging reduces prediction variance without requiring retraining.

Fig. 2 shows the 8 D4 transforms applied to a real lesion image from the training set.

D. Image Data Pipeline

Images are stored as JPEG-compressed byte arrays in HDF5, decoded on-the-fly via OpenCV to avoid disk I/O bottlenecks from 401K individual files. Each DataLoader worker opens its own HDF5 file handle (via a custom `worker_init_fn`) to avoid Python GIL contention and ensure thread-safe access.

VII. Phase 3: Stacking Meta-Learner

Phase 3 combines predictions from Phase 1 (tabular) and Phase 2 (four image models) through a meta-learner trained on OOF predictions.

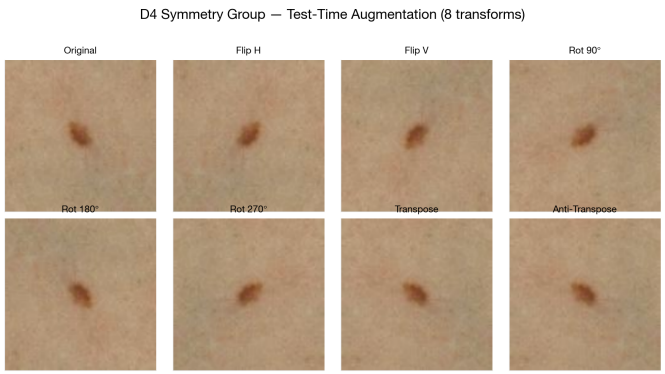

Fig. 2: D4 symmetry group applied to a real lesion from the training set. All 8 transforms (4 rotations $\times$ 2 reflections) are averaged at test time via TTA (Eq. 11).

A. Feature Construction

Each model's OOF predictions are rank-transformed to percentile scores in $[0, 1]$:

$$r_m(i) = \frac{\text{rank}(\hat{p}_m(i))}{N} \tag{12}$$

This produces five features: `rank_tabular`, `rank_effnet`, `rank_eva02`, `rank_convnext`, `rank_swin`. Rank transformation is essential because the five base models produce predictions on different scales (GBDT rank-percentiles, sigmoid probabilities), and raw values would give undue weight to whichever scale has larger magnitude.

B. Stacking Methods

We evaluate four meta-learner approaches with increasing model complexity:

a) (1) Rank ensemble (equal weights): Simple average of all five rank features, reusing the `RankEnsemble` from Phase 1. Zero overfitting risk since no parameters are learned.

b) (2) Rank ensemble (pAUC-weighted): Weighted average where each model's weight equals its individual pAUC score, giving more influence to stronger base models.

c) (3) Logistic regression: `LogisticRegression(class_weight=balanced, C=1.0, solver=lbfgs)` from scikit-learn, trained per CV fold on rank features. The linear decision boundary and L2 regularisation provide an appropriate inductive bias for only 5 features and 393 positives — complex nonlinear models cannot be reliably estimated with this positive count.

d) (4) Conservative LightGBM: `n_estimators=500`, `lr=0.01`, `num_leaves=7`, `max_depth=3`, `min_child_samples=50`, L_1=L_2=1.0, `scale_pos_weight=100`, 50-round early stopping, averaged over 3 seeds. Despite heavy regularisation, the GBDT overfits — confirming that with only 5 features and $\sim$393 positive samples, tree-based nonlinearity is counterproductive.

TABLE V: LightGBM Per-Fold OOF Metrics (5-fold CV, seed avg $\times$5).

Fold	pAUC	AUC	Brier	ECE
1	0.1408	0.9230	0.0639	0.1210
2	0.1343	0.9140	0.0579	0.1112
3	0.1459	0.9276	0.0411	0.0946
4	0.1453	0.9170	0.0841	0.1338
5	0.1337	0.9175	0.0061	0.0263
Mean	**0.1400**	**0.9198**	0.0506	0.0974
Std	0.0052	0.0049	—	—

TABLE VI: XGBoost Per-Fold OOF Metrics.

Fold	pAUC	AUC	Brier	ECE
1	0.1653	0.9584	0.00199	0.00722
2	0.1668	0.9608	0.00185	0.00631
3	0.1690	0.9623	0.00369	0.01424
4	0.1848	0.9792	0.00100	0.00045
5	0.1653	0.9580	0.00377	0.01518
Mean	**0.1703**	**0.9637**	0.00246	0.00868
Std	0.0074	0.0079	—	—

TABLE VII: CatBoost Per-Fold OOF Metrics.

Fold	pAUC	AUC	Brier	ECE
1	0.1731	0.9674	0.00164	0.00793
2	0.1626	0.9559	0.00224	0.01429
3	0.1655	0.9585	0.00208	0.01365
4	0.1746	0.9688	0.00187	0.01343
5	0.1577	0.9482	0.00181	0.00982
Mean	**0.1667**	**0.9598**	0.00193	0.01182
Std	0.0064	0.0076	—	—

TABLE VIII: SVM Baseline Per-Fold OOF Metrics (20K subsample).

Fold	pAUC	AUC	Brier	ECE
1	0.1160	0.8830	0.000971	0.000112
2	0.1347	0.9105	0.000961	0.000124
3	0.1464	0.9270	0.001029	0.000214
4	0.1120	0.8725	0.000950	0.000018
5	0.0745	0.8329	0.000952	0.000243
Mean	**0.1167**	**0.8852**	0.000973	0.000142
Std	0.0245	0.0325	—	—

VIII. Experiments and Results

A. Phase 1: Tabular Results

Tables V–VIII report per-fold metrics for all tabular models.

B. Phase 1: Summary

Table IX summarises all Phase 1 systems.

C. Phase 1: Calibration Analysis

Isotonic regression reduces ECE from 0.499 to effectively zero while pAUC slightly increases (+0.0019). Fig. 3 shows reliability diagrams: the raw ensemble is severely miscalibrated

TABLE IX: Phase 1 Summary. Target: pAUC ≥ 0.16.

Model	pAUC (mean $\pm$ std)	AUC	Brier	ECE
LightGBM	0.1400 ± 0.0052	0.920	0.0506	0.0974
XGBoost	$\mathbf{0.1703 \pm 0.0074}$	0.964	0.00246	0.00868
CatBoost	0.1667 ± 0.0064	0.960	0.00193	0.01182
SVM (baseline)	0.1167 ± 0.0245	0.885	0.000973	0.000142
GBDT Ensemble	0.1653	0.956	$0.299^{\dagger}$	$0.499^{\dagger}$
Ens. + Isotonic	**0.1672**	0.958	**0.000907**	≈ 0

† Rank-percentile outputs are not calibrated probabilities; Brier/ECE are meaningless before calibration.

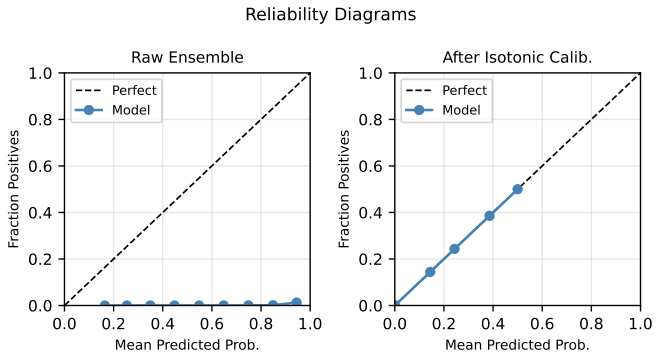

Fig. 3: Reliability diagrams from OOF predictions. Left: raw rank-ensemble (miscalibrated). Right: after isotonic regression (near-perfect calibration, ECE ≈ 0).

TABLE X: Phase 2 Image Model Results (5-Fold CV, TTA).

Model	Params	Input	pAUC	AUC
EfficientNetV2-S	20M	224^2	0.1399	0.9239
EVA02-Small	22M	336^2	0.1403	0.9220
ConvNeXtV2-B	88M	224^2	0.1464	0.9286
SwinV2-B	**87M**	$\mathbf{256^2}$	**0.1549**	**0.9388**

(rank-percentile scores are not probabilities) while calibration brings the curve onto the diagonal.

D. Phase 1: ROC Curve Analysis

Fig. 4 plots the full ROC curves from OOF predictions. The shaded region (FPR ≤ 0.20) is the pAUC scoring zone. Fig. 5 zooms into this critical region, showing XGBoost and CatBoost leading in the high-sensitivity zone.

Fig. 6 shows the score distributions of each model for benign vs. malignant lesions.

E. Phase 2: Image Model Results

Table X summarises all image backbone results. Table XI provides the per-fold breakdown.

Fold 3 is consistently the hardest across all models and both phases, suggesting genuinely difficult cases in that patient split rather than model-specific weakness. Larger models (87–88M params) consistently outperform smaller ones (20–22M), indicating that the visual signal in skin lesion images benefits from greater model capacity.

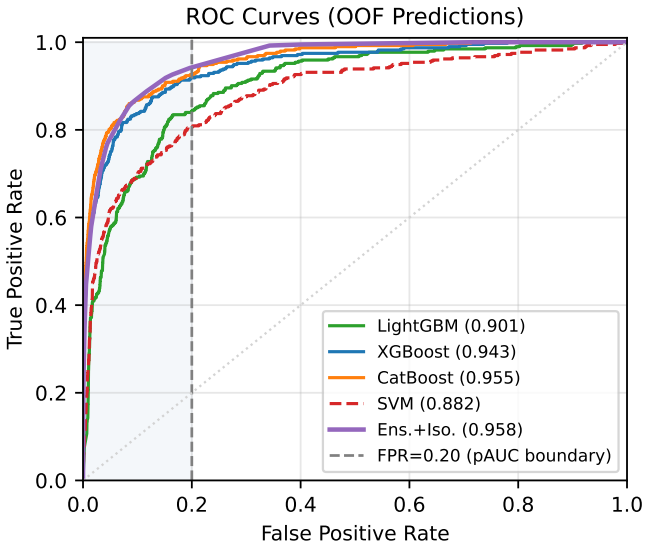

Fig. 4: Phase 1 ROC curves from OOF predictions. Shaded region (FPR ≤ 0.20) is the pAUC scoring zone.

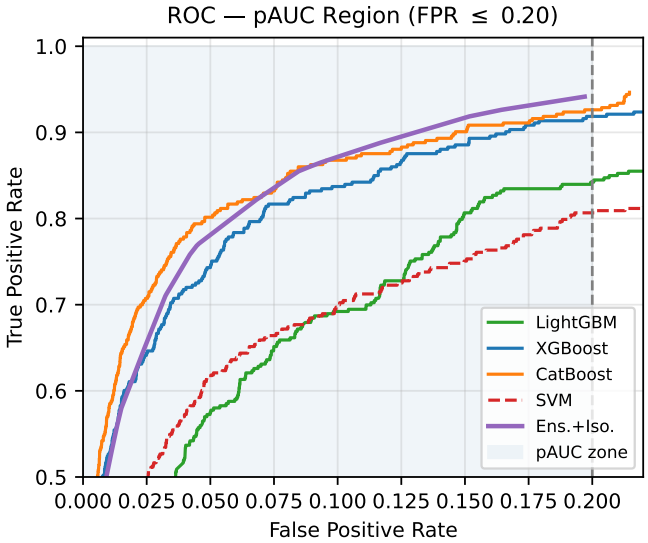

Fig. 5: Zoom into the pAUC region (FPR ≤ 0.20, TPR ≥ 0.50). XGBoost and CatBoost lead in this high-sensitivity zone.

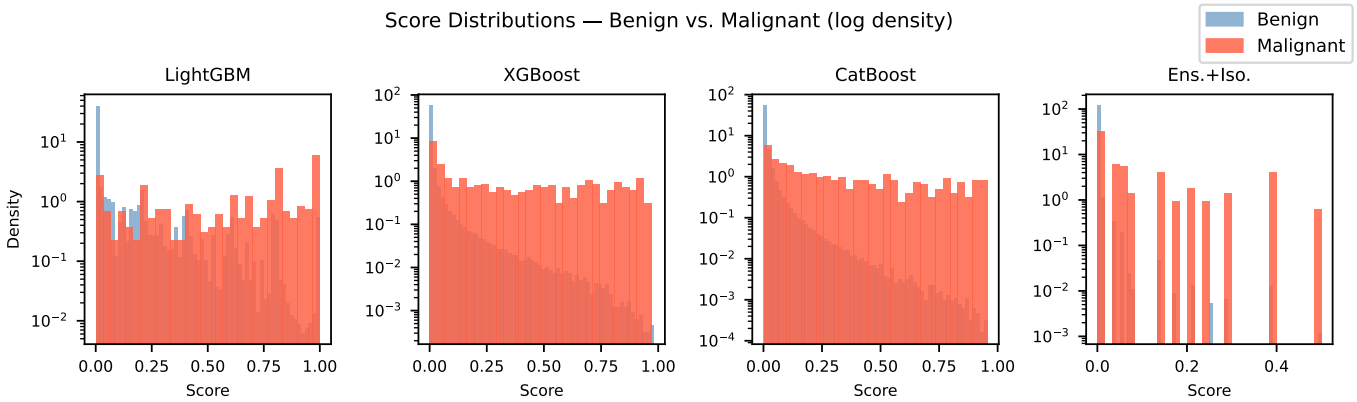

Fig. 6: Score distributions (log density) for benign (blue) and malignant (red) lesions from OOF predictions. Models left to right: LightGBM, XGBoost, CatBoost, Ens.+Iso.

TABLE XI: Phase 2 Per-Fold pAUC Breakdown.

Fold	EffNetV2	EVA02	ConvNeXt	SwinV2
0	0.1383	0.1502	0.1519	0.1529
1	0.1528	0.1557	0.1513	0.1650
2	0.1551	0.1704	0.1705	0.1807
3	0.1106	0.1321	0.1334	0.1368
4	0.1671	0.1669	0.1668	0.1668
Mean	0.1399	0.1403	0.1464	**0.1549**

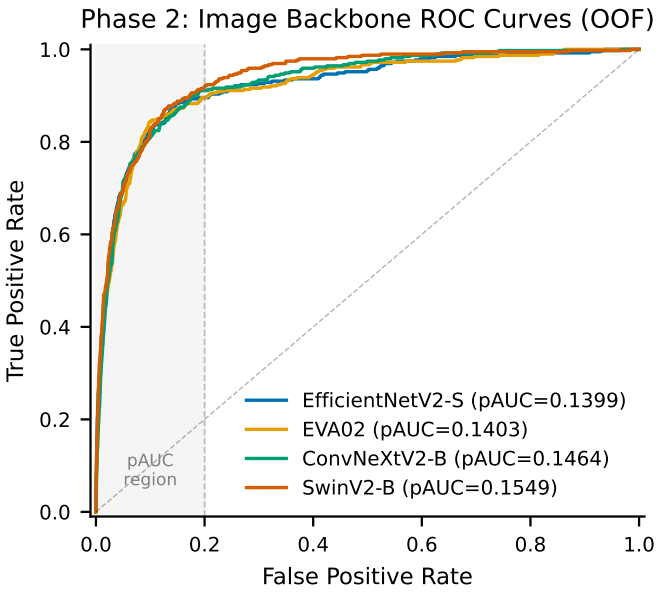

Fig. 7: Phase 2 ROC curves from OOF predictions. SwinV2-B achieves the best separation in the pAUC scoring zone (shaded).

Fig. 7 shows ROC curves for all four image backbones.

F. Phase 3: Stacking Results

Table XII compares all stacking methods.

The logistic regression stacker achieves the best pAUC of **0.1745**, a 5.6% improvement over the best individual model

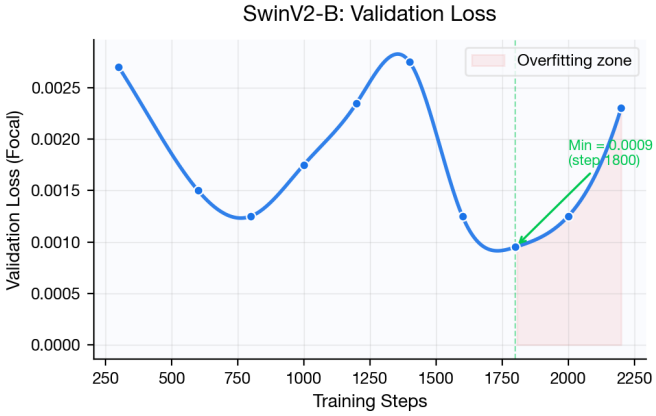

Fig. 8: SwinV2-B validation loss (focal) across training steps (data from Weights & Biases). Minimum at step ∼1.8K; subsequent rise triggers early stopping. Shaded region indicates overfitting zone.

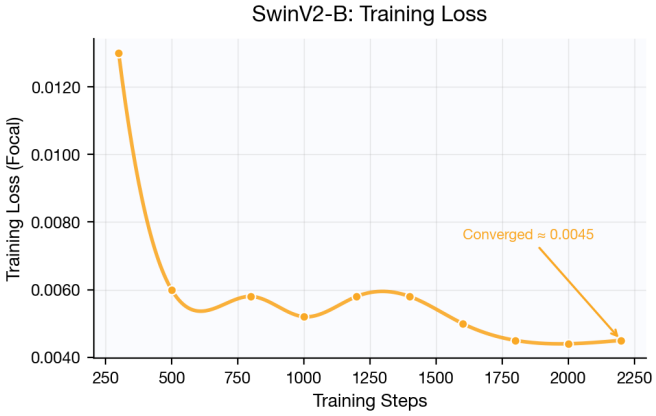

Fig. 9: SwinV2-B training loss decreases monotonically from 0.013 to 0.004, confirming the model is learning without memorisation.

TABLE XII: Phase 3 Stacking Comparison.

Method	pAUC	Overfit Risk
Individual models (rank-transformed):		
Tabular (Phase 1)	0.1653	—
EfficientNetV2-S	0.1399	—
EVA02-Small	0.1403	—
ConvNeXtV2-B	0.1464	—
SwinV2-B	0.1549	—
Stacking methods:		
Rank ens. (equal)	0.1708	Zero
Rank ens. (pAUC-wt.)	0.1715	Zero
LogReg stacker	**0.1745**	Very low
LightGBM stacker	0.1645	Low (overfit)

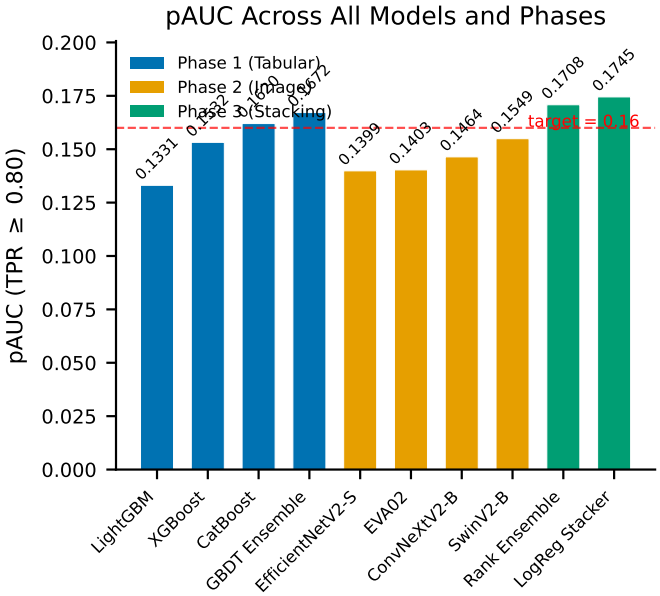

Fig. 10: pAUC comparison across all phases. Red dashed line indicates the Phase 1 target (0.16). The LogReg stacker (Phase 3) achieves the best overall pAUC of 0.1745.

(tabular, 0.1653). The LightGBM stacker *underperforms* the simple rank ensemble despite heavier regularisation, confirming that with only 393 positive samples and 5 features, tree-based nonlinearity is counterproductive.

Fig. 10 shows the pAUC comparison across all three phases.

Fig. 11 visualises the progression from individual models to the stacked ensemble.

Fig. 12 compares all four stacking methods.

G. Cross-Phase Comparison

Table XIII presents the final results across all three phases.

IX. Discussion

A. Why Tabular Outperforms Image Individually

Phase 1 tabular models (pAUC = 0.1653–0.1672) outperform all four image models individually (pAUC = 0.1399–0.1549). This is expected for two reasons: (1) the `tbp_lv_*` metadata columns are *quantitative measurements* of the ABCDE criteria,

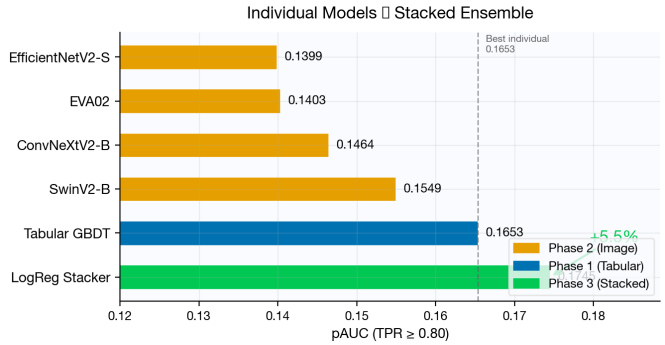

Fig. 11: Individual model pAUC vs. LogReg stacker. Image models individually underperform the tabular baseline, but stacking all five sources yields a +5.6% improvement.

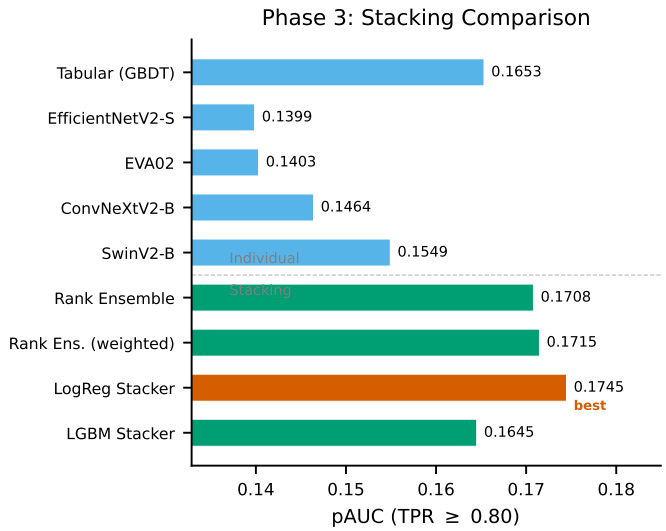

Fig. 12: Phase 3 stacking method comparison. LogReg outperforms both rank ensembles and the LightGBM stacker, which overfits with only 393 positive samples.

TABLE XIII: Grand Summary Across All Phases.

Phase	Best Model	pAUC
1 (Tabular)	Calibrated GBDT Ensemble	0.1672
2 (Image)	SwinV2-B	0.1549
3 (Stacking)	**LogReg on ranks**	**0.1745**

while image models must learn to extract these features implicitly from pixel data; (2) the ugly duckling z-scores provide patient-relative context that a single-image model cannot access. However, image models capture visual patterns (subtle texture, vascular structures, irregular pigment networks) not represented in the metadata, as evidenced by the +5.6% pAUC gain from stacking.

B. Architecture Diversity and Stacking Gains

The stacking improvement (0.1653 → 0.1745) confirms that tabular and image models make complementary errors.

Within image models, the four architectures also contribute differently: SwinV2-B's shifted-window attention captures multi-scale spatial relationships, while ConvNeXtV2-B's strong local priors excel at texture features. Even the weaker models (EfficientNetV2-S, EVA02-Small) contribute to the stacking gain because they make *different* errors on different subsets of samples.

C. Why LogReg Beats LightGBM as Meta-Learner

With only 393 positive samples and 5 input features, the optimal meta-learner is the simplest one that can capture the signal. Logistic regression's linear decision boundary has $\mathcal{O}(d)$ parameters (5 weights + bias = 6 parameters), while LightGBM with `num_leaves=7` has $\mathcal{O}(7 \times \text{depth})$ effective parameters per tree $\times$ 500 trees. The gap between `lgbm_stacker` (0.1645) and `logreg_stacker` (0.1745) is a clear case of the bias-variance trade-off: the GBDT's additional capacity *hurts* because there is insufficient positive data to estimate nonlinear interactions reliably.

D. LightGBM vs. XGBoost Imbalance Semantics

The 0.0486 pAUC gap between LightGBM (0.1217) and XGBoost (0.1703) in early experiments reflects a fundamental API difference. LightGBM's `min_child_samples` imposes a count floor that ignores the boosted weight of positive samples. Under `scale_pos_weight=100`, a single positive sample contributes $h \approx 100 \times p(1-p) \approx 25$ hessian weight (at $p \approx 0.5$), satisfying XGBoost's `min_child_weight=10` immediately.

E. Fold 3 Difficulty

Fold 3 is consistently the hardest fold across all models in both Phase 1 and Phase 2 (Tables V–XI). Since folds are grouped by patient, this suggests that the patients in fold 3 have lesions that are inherently harder to classify — possibly due to atypical presentations or lower-quality imaging. This consistent pattern across architecturally diverse models rules out model-specific bias as the cause.

F. Limitations and Future Work

- **Hyperparameter search**: GBDT and image model hyperparameters were set manually; Bayesian optimisation (Optuna) could yield +0.01–0.02 pAUC.
- **Higher resolution**: Image models used 224–336px; higher resolution (384–512px) may capture fine-grained dermoscopic structures.
- **Advanced augmentation**: MixUp [30] or CutMix could improve image model generalisation.
- **Patient-level ugly duckling on predictions**: Computing rank deviation of each lesion's image prediction from the patient mean could provide stacking features that capture the ugly duckling signal at the prediction level.
- **Neural tabular models**: TabNet or FT-Transformer could complement GBDTs in the tabular phase.

X. Conclusion

We developed a three-phase pipeline for skin lesion classification on the ISIC 2024 SLICE-3D dataset, progressing from tabular features (pAUC = 0.1672) through image deep learning (pAUC = 0.1549 best individual) to a stacking meta-learner achieving **pAUC = 0.1745**.

Key findings: (1) Tabular metadata with ABCDE feature engineering and ugly duckling z-scores outperforms image models individually, because the `tbp_lv_*` columns *are* quantitative ABCDE measurements. (2) Image models capture complementary visual signals: stacking improves pAUC by 5.6% over the best individual model. (3) Architecture diversity (CNN, ViT, ConvNet, hybrid) matters more than single-model optimisation for stacking ensembles. (4) With only 393 positives, logistic regression outperforms LightGBM as a meta-learner — a textbook bias-variance trade-off outcome. (5) LightGBM's `min_child_samples` is semantically incompatible with `scale_pos_weight` for imbalanced data. (6) Identical CV splits across all phases enable leak-free OOF alignment, which is the foundation that makes multi-phase stacking possible.

References

[1] International Skin Imaging Collaboration, "ISIC 2024 Challenge: Skin Lesion Classification," https://www.kaggle.com/competitions/isic-2024-challenge, 2024.

[2] G. Ke *et al.*, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," *NeurIPS*, vol. 30, 2017.

[3] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *Proc. 22nd ACM SIGKDD*, pp. 785–794, 2016.

[4] L. Prokhorenkova *et al.*, "CatBoost: Unbiased Boosting with Categorical Features," *NeurIPS*, vol. 31, 2018.

[5] C. Cortes and V. Vapnik, "Support-Vector Networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[6] J. Platt, "Probabilistic Outputs for Support Vector Machines," *Advances in Large Margin Classifiers*, pp. 61–74, 1999.

[7] A. Niculescu-Mizil and R. Caruana, "Predicting Good Probabilities with Supervised Learning," *Proc. ICML*, pp. 625–632, 2005.

[8] J. J. Grob and J. J. Bonerandi, "The 'Ugly Duckling' Sign," *Archives of Dermatology*, vol. 134, no. 1, pp. 103–104, 1998.

[9] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.

[10] D. K. McClish, "Analyzing a Portion of the ROC Curve," *Medical Decision Making*, vol. 9, no. 3, pp. 190–195, 1989.

[11] R. L. Siegel *et al.*, "Cancer Statistics, 2024," *CA: A Cancer Journal for Clinicians*, vol. 74, no. 1, pp. 12–49, 2024.

[12] R. J. Friedman, D. S. Rigel, and A. W. Kopf, "Early Detection of Malignant Melanoma: The Role of Physician Examination and Self-Examination of the Skin," *CA: A Cancer Journal for Clinicians*, vol. 35, no. 3, pp. 130–151, 1985.

[13] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" *NeurIPS*, 2022.

[14] N. Codella *et al.*, "Skin Lesion Analysis Toward Melanoma Detection: A Challenge at ISBI 2017," *Proc. IEEE ISBI*, pp. 168–172, 2018.

[15] B. Zadrozny and C. Elkan, "Transforming Classifier Scores into Accurate Multiclass Probability Estimates," *Proc. ACM KDD*, pp. 694–699, 2002.

[16] C. Guo *et al.*, "On Calibration of Modern Neural Networks," *Proc. ICML*, pp. 1321–1330, 2017.

[17] M. Tan and Q. Le, "EfficientNetV2: Smaller Models and Faster Training," *Proc. ICML*, pp. 10096–10106, 2021.

[18] A. Dosovitskiy *et al.*, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," *Proc. ICLR*, 2021.

[19] Y. Fang *et al.*, "EVA-02: A Visual Representation for Neon Genesis," *arXiv preprint arXiv:2303.11331*, 2023.

[20] Z. Liu *et al.*, "Swin Transformer V2: Scaling Up Capacity and Resolution," *Proc. CVPR*, pp. 12009–12019, 2022.
[21] S. Woo *et al.*, "ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders," *Proc. CVPR*, pp. 16133–16142, 2023.
[22] T.-Y. Lin *et al.*, "Focal Loss for Dense Object Detection," *Proc. ICCV*, pp. 2980–2988, 2017.
[23] R. Wightman, "PyTorch Image Models," https://github.com/huggingface/pytorch-image-models, 2019.
[24] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," *Proc. ICLR*, 2019.
[25] L. N. Smith and N. Topin, "Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates," *Proc. SPIE*, vol. 11006, 2019.
[26] O. Russakovsky *et al.*, "ImageNet Large Scale Visual Recognition Challenge," *IJCV*, vol. 115, no. 3, pp. 211–252, 2015.
[27] B. Cassidy *et al.*, "Analysis of the ISIC Image Datasets: Usage, Benchmarks and Recommendations," *Medical Image Analysis*, vol. 75, 102305, 2022.
[28] D. H. Wolpert, "Stacked Generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.
[29] R. Caruana *et al.*, "Ensemble Selection from Libraries of Models," *Proc. ICML*, pp. 18–25, 2004.
[30] H. Zhang *et al.*, "mixup: Beyond Empirical Risk Minimization," *Proc. ICLR*, 2018.